

generative ai on the environment even when local power is generated differently?

Posted by u/Lost-Sheepherder7413 • 0 points • 9 comments

posting this because someone posed this as a counterargument to my worries of generative AI on the environment and i dont know enough to refute it. i was told that in a state where our energy is generated by water-power, it doesn't matter if we use AI, since our power is coming from here? this sounds incredibly misinformed to me but i honestly just don't know enough about power systems to understand why or how it's wrong. can someone explain further?

Comments

u/Human_certified • 2 points • 2025-02-12 19:08 UTC

Just gonna leave this study here again:

[<https://andymasley.substack.com/p/individual-ai-use-is-not-bad-for>]
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TL;DR - don't worry about AI, it's a drop in the ocean, especially compared to some really impactful things you can do.

u/DarkJayson • 3 points • 2025-02-12 12:35 UTC

Here you go someone did the math here, this is the results <https://blog.kyleggiero.me/image-generators-energy-usage/>

Here is also a break down of his data.

<https://preview.redd.it/wdkhrpb9dpie1.jpeg?width=2580&format=pjpg&auto=webp&s=b48414366d173f083a520e364b2995a22c7f9d47>

u/AppearanceHeavy6724 • 2 points • 2025-02-12 10:11 UTC

if you drive a gasoline car, 1 minute of ride is equal a whole day of using chatgpt.

u/PowderMuse • 3 points • 2025-02-12 09:42 UTC

If you do 20 prompts an hour that's 58Wh per hour.

Compare that to: a refrigerator 62, a TV 150, a gaming PC 250, an electric oven 2500.

AI is one of the things in your life that is fairly low energy but arguably one of the most beneficial.

u/Primary_Spinach7333 • 1 points • 2025-02-14 04:28 UTC

If these people are still making these kind of stupid points even after you give data because they can't do math or something, then I don't think this'll convince them, pal. It's not you, it's them

Ok but in all seriousness thank you for this piece of information

u/PowderMuse • 2 points • 2025-02-14 06:16 UTC

Actually since I posted that, I did some research and the latest models use less than a third of what I said at 0.9 Wh per prompt.

u/OverCategory6046 • 12 points • 2025-02-12 01:35 UTC

AI uses the power of wherever the server farm is located, not your local power.

Also as far as energy use goes, theres farrrr worse industries to go for other than AI.. same for water use, golf courses use a HUGE amount more water.

u/MotorStrict8568 • 1 points • 2025-02-14 14:45 UTC

This is true, but I believe data centers consume about 1% of the USA's energy, which is not insignificant. And that number is likely rising with the proliferation of AI. That's one reason some companies are looking into nuclear power.

u/MysteriousPepper8908 • 12 points • 2025-02-12 01:40 UTC

Unless you generate locally, then it just uses however much energy your computer can output depending on how much its taxing your system. Roughly about as much as you'd use playing a game or doing some rendering.

u/OverCategory6046 • 3 points • 2025-02-12 01:46 UTC

Indeed, but `\*most\*` people aren't using it locally, in which case yea.

Might use a little bit less than running a game maxxed out? Not sure how it pegs the CPU, never checked when I've ran stuff locally

u/MysteriousPepper8908 • 4 points • 2025-02-12 01:54 UTC

There's a huge amount of variance depending on the game, resolution, tons of stuff but the main idea is that your computer can only draw so much power based on its power supply so there's an upper limit there. Most people aren't generating locally but more than a year ago, I'd bet, and I'd expect that to keep going up as we start to see models that are good enough able to run locally on your average phone. Deepseek can run on phones, I guess, but I think the IQ drops fairly substantially with that level of quantization.

u/Hugglebuns • 5 points • 2025-02-12 01:33 UTC

Basically cloud servers want to save money on electricity, hydropower is the cheapest per watt-hour (most green is cheaper than fossils nowadays), and so a lot of cloud computing is built out in the rural zones near mountain ranges to tap into the hydro power

It also helps that hydro has a negligible carbon footprint and if the main problem with energy use is carbon footprint, then voila.